

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{18} \tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \text{LF}_{2,2,0} [m_d^r, m_q^p] \delta_{i1i2} - \frac{1}{36} \tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \text{LF}_{3,2,-1} [m_d^r, m_q^p] \delta_{i1i2} + \right. \\
& \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \text{LF}_{2,2,0} [m_e^r, m_l^p] \delta_{i1i2} - \frac{1}{36} \tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \text{LF}_{3,2,-1} [m_e^r, m_l^p] \delta_{i1i2} - \\
& \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \text{LF}_{3,1,0} [m_l^p, m_e^r] \delta_{i1i2} - \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \text{LF}_{3,2,-1} [m_l^p, m_e^r] \delta_{i1i2} + \\
& \frac{1}{8} \tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \text{LF}_{4,1,-1} [m_l^p, m_e^r] \delta_{i1i2} - \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \text{LF}_{5,1,-2} [m_l^p, m_e^r] \delta_{i1i2} - \\
& \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \text{LF}_{3,1,0} [m_q^p, m_d^r] \delta_{i1i2} - \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \text{LF}_{3,2,-1} [m_q^p, m_d^r] \delta_{i1i2} + \\
& \frac{5}{24} \tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \text{LF}_{4,1,-1} [m_q^p, m_d^r] \delta_{i1i2} - \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \text{LF}_{5,1,-2} [m_q^p, m_d^r] \delta_{i1i2} - \\
& \frac{1}{36} \tilde{\mu} g_1^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{2,2,0} [m_q^p, m_u^r] \delta_{i1i2} + \frac{1}{72} \tilde{\mu} g_1^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{3,2,-1} [m_q^p, m_u^r] \delta_{i1i2} + \\
& \frac{1}{36} \tilde{\mu} g_1^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{3,1,0} [m_u^r, m_q^p] \delta_{i1i2} + \frac{1}{36} \tilde{\mu} g_1^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{3,2,-1} [m_u^r, m_q^p] \delta_{i1i2} + \\
& \frac{1}{12} \tilde{\mu} g_1^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{4,1,-1} [m_u^r, m_q^p] \delta_{i1i2} - \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_u}^{pr} a_u^{pr} \text{LF}_{5,1,-2} [m_u^r, m_q^p] \delta_{i1i2} + \\
& \frac{1}{24} m_1 \tilde{\mu} g_1^4 \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] \delta_{i1i2} - \frac{1}{18} m_1 \tilde{\mu} g_1^4 \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] \delta_{i1i2} + \\
& \frac{1}{8} m_2 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{6} m_2 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] \delta_{i1i2} + \frac{1}{72} m_1 \tilde{\mu} g_1^2 \\
& \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,0} [m_1, m_d^p, m_q^{i2}] + \frac{1}{144} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1} [m_1, m_d^p, m_q^{i2}] - \\
& \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{3,1,1,-1} [m_1, m_d^p, m_q^{i2}] + \frac{1}{12} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,0} [m_1, m_d^p, \tilde{\mu}] - \\
& \frac{1}{12} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{3,1,1,-1} [m_1, m_d^p, \tilde{\mu}] - \frac{1}{144} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1} [m_1, m_q^{i2}, m_d^p] + \\
& \frac{1}{36} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,0} [m_1, m_q^{i2}, m_u^p] - \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \\
& \text{LF}_{2,2,1,-1} [m_1, m_q^{i2}, m_u^p] - \frac{1}{36} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{3,1,1,-1} [m_1, m_q^{i2}, m_u^p] + \\
& \frac{1}{72} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1} [m_1, m_u^p, m_q^{i2}] - \frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,0} [m_1, m_u^p, \tilde{\mu}] + \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{3,1,1,-1} [m_1, m_u^p, \tilde{\mu}] - \frac{5}{48} m_1 \tilde{\mu} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^p] - \\
& \frac{1}{48} m_1 \tilde{\mu} g_1^2 (\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}) \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i2}] + \\
& \frac{7}{48} m_1 \tilde{\mu} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^p] - \frac{3}{16} m_2 \tilde{\mu} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_d^p] - \\
& \frac{3}{16} m_2 \tilde{\mu} g_2^2 (\overline{y_d}^{i2p} y_d^{i1p} - \overline{y_u}^{i2p} y_u^{i1p}) \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i2}] + \\
& \frac{3}{16} m_2 \tilde{\mu} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_u^p] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,1,1,0} [m_3, m_d^p, m_q^{i2}] - \\
& \frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1} [m_3, m_d^p, m_q^{i2}] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{3,1,1,-1} [m_3, m_d^p, m_q^{i2}] + \\
& \frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} \text{LF}_{2,2,1,-1} [m_3, m_q^{i2}, m_d^p] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,1,1,0} [m_3, m_q^{i2}, m_u^p] - \\
& \frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1} [m_3, m_q^{i2}, m_u^p] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{3,1,1,-1} [m_3, m_q^{i2}, m_u^p] + \\
& \frac{1}{6} m_3 \tilde{\mu} g_3^2 \overline{y_u}^{i2p} y_u^{i1p} \text{LF}_{2,2,1,-1} [m_3, m_u^p, m_q^{i2}] - \frac{1}{16} \tilde{\mu} \overline{y_d}^{i2p} \overline{y_u}^{rs} y_u^{i1s} a_d^{rp} \\
& \text{LF}_{2,2,1,-1} [m_d^p, \tilde{\mu}, m_q^r] - \frac{1}{16} \tilde{\mu} \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} a_u^{ps} \text{LF}_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_u^s] + \\
& \frac{1}{16} \tilde{\mu} \overline{y_d}^{i2p} \overline{y_u}^{rs} y_u^{i1s} a_d^{rp} \text{LF}_{2,2,1,-1} [m_q^r, \tilde{\mu}, m_d^p] + \frac{1}{16} \tilde{\mu} \overline{y_d}^{pr} y_d^{i1r} \overline{y_u}^{i2s} a_u^{ps} \\
& \text{LF}_{2,2,1,-1} [m_u^s, \tilde{\mu}, m_q^p] + \frac{1}{24} m_1 \tilde{\mu} g_1^2 (\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}) \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_q^{i2}] - \\
& \frac{1}{24} m_1 \tilde{\mu} g_1^2 (\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}) \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_q^{i2}] + \\
& \frac{3}{8} m_2 \tilde{\mu} g_2^2 (\overline{y_d}^{i2p} y_d^{i1p} - \overline{y_u}^{i2p} y_u^{i1p}) \text{LF}_{2,1,1,0} [\tilde{\mu}, m_2, m_q^{i2}] + \\
& \frac{3}{8} m_2 \tilde{\mu} g_2^2 (-\overline{y_d}^{i2p} y_d^{i1p} + \overline{y_u}^{i2p} y_u^{i1p}) \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_2, m_q^{i$$